ABEE2007

Environmental Science for Architects
Spring Semester Project Submission

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By submitting this work I accept that it is entirely my group's work and that any external material has been fully attributed to the original author(s) / sources

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Task 2: Reflect on and critically appraise the environmental design of the existing house – Pre retrofit SWOT analysis

Building element U-Values

Building element	U-value (W/m^2K)	Document L1 Table 1.1	LETI Document
Wall	0.576	0.18	0.13-0.15
Roof	0.666	0.11	0.10-0.12
Floor	0.600	0.13	0.08-0.10
Windows	2.800	1.2	0.80 triple glazing
Doors	2.500	1	1.00

Walls

A strength would be that the wall is constructed with both its inner and outer leaves made of brick. Brick has a low thermal conductivity, which means it limits heat transfer through the building envelope, contributing to a lower U- value. However, brick alone offers less structural strength compared to a wall made from a combination of brick and concrete block. Replacing the inner leaf with concrete blocks would enhance the structural integrity of the wall. That said, concrete has a higher thermal conductivity than brick, which increases the rate of heat transfer through the envelope.

The wall also includes a 20mm cavity between the inner and outer leaves. This cavity traps air, reducing heat conduction and thereby slowing down heat loss, hence a strength. However, due to the absence of insulation, the wall has a relatively high U-value of 0.576 W/m²K and a Heat Loss Coefficient (HLC) of 46.468 W/K, which allows more heat to escape through the building envelope, making it a weakness. Compared to the LETI guidance range of 0.13–0.15 W/m²K, it's clear that the current U-value is significantly higher.

As an opportunity to reduce the U-value, insulation should be incorporated into the wall's structure, which would slow down heat transfer. However, this would increase the wall's overall thickness and reduce the total usable floor area.

Roof

Part L requires a maximum roof U-value of 0.11 W/m²K; however, the current roof has a U-value of 0.666 W/m²K, which fails to meet this requirement, causing the roof a weakness in design. Among all building elements, the pitched roof has the highest U-value, largely due to its construction and the natural tendency of warm air to rise.

The elevated U-value is primarily caused by insufficient insulation (only 50mm of wool) placed between the joists. This setup leads to thermal bridging, where heat escapes easily through the uninsulated joist areas. The limited thickness of plaster, plasterboard, and insulation further reduces the roof's thermal resistance, contributing to greater heat loss.

The existing cold roof design, with insulation located at ceiling level between joists, intensifies thermal bridging due to breaks in the insulation layer. As the sun heats the roof surface, the air around it warms up, rises, and carries heat away. Conversely, during colder nights, denser cool air can sink through the roof structure into the living space, increasing heat loss. As a result, the roof's U-value is 0.666 W/m²K, with an associated HLC of 55.150 W/K which is significantly higher than the LETI guidance range of 0.10–0.12 W/m²K.

To address this, an additional insulation layer should be installed above the joists to create a continuous thermal barrier and form a warm roof, reducing thermal bridging. Moreover, ventilating the attic space between the ceiling and the roof is essential to prevent moisture build-up and interstitial condensation, which could pose health risks to occupants.

Floor

Approved Document Part L requires a floor U-value of 0.13 W/m²K, yet the current floor achieves a U-value of 0.60 W/m²K, indicating non-compliance, hence a weakness. The concrete floor allows considerable heat loss to the ground due to the absence of insulation.

Two main factors contribute to this high U-value. Lack of insulation is the first one, which results in no thermal resistance and enables unrestricted heat flow. Floor construction is the second contributing factor, comprising 50mm concrete screed over 100mm oversite concrete. While concrete acts as a thermal mass (absorbing and storing heat) it offers limited insulation value and does little to reduce the U-value on its own.

Although concrete has a relatively low thermal conductivity compared to some materials, the absence of insulation leads to a high U-value of 0.600 W/m²K and an HLC of 55.150 W/K. This is significantly above the recommended range of 0.08–0.10 W/m²K stated in the LETI guidance.

To improve thermal performance, insulation should be incorporated into the floor construction. This would increase the floor's thickness and resistance, slowing down the rate of heat transfer and reducing the overall U- value.

Windows and Doors

The U-values of both the windows (2.8 W/m²K) and doors (2.5 W/m²K) significantly exceed the limits set by Approved Document Part L and LETI guidelines. This poor performance is primarily due to the lack of triple glazing in the windows, making them highly susceptible to heat loss. Similarly, the doors feature single-glazed glass and are constructed from timber, with the single glazing being a major contributor to their high U-value.

In addition to thermal inefficiency, the doors may also allow cold air to enter through gaps, potentially leading to excessive ventilation and further heat loss. To enhance the thermal performance of both elements, upgrading to thicker glazing (such as double or triple glazing) would substantially reduce heat transfer and improve compliance with energy efficiency standards

Area of windows

Elevation	North	East	South	West
Total Window Area	0	9.858	2.64	4.945
Total Wall Area	17.133	22.067	14.493	26.98
Percentage of Window over Wall area (%)	0	44.673	18.216	18.328
Percentage of Window over Wall area (%) LETI	10-15	10-15	20-25	10-15

When comparing the window-to-wall ratios of the existing house to the LETI recommendations, it becomes clear that the house does not comply. On the north elevation, there are no windows, resulting in a 0% window-to-wall ratio, significantly below LETI's suggested 10–15%. This likely limits natural daylight in rooms on that side.

In contrast, the west and east elevations have window areas of 18.328% and 44.673% respectively, both exceeding LETI's recommended range. Excessive glazing on these elevations can result in overheating and glare, reducing occupant comfort. To address this, the window areas could be reduced or selectively removed where appropriate.

The south elevation, with a window area of 18.216%, also is under LETI's recommended range of 20–25%, indicating a low level of daylighting on that side of the house, hence another weakness of the design.

Ventilation

Type of Ventilation	Before retrofit	Part F calculation	Part F criteria	LETI criteria
Natural	1.1ach-1	0.285ach-1	5 m3/h.m^2 @50PA	1 m3/h. m2@50 PA.
Mechanical bathroom	0.6389 l/s	1.513 l/s	/	90% heat recovery
Mechanical Kitchen	5 l/s	3.717 l/s	/	

Natural ventilation:

As discussed above, the building envelope demonstrates certain strengths; however, its weaknesses significantly affect the effectiveness of natural ventilation. According to the LETI guidelines for small-scale housing archetypes, air tightness should be below 1 $m^3/h \cdot m^2$ at 50 Pa. In contrast, the current air permeability of the house is 1 $m^3/h \cdot m^2$ at 50 Pa, which exceeds the recommended threshold. This indicates that the envelope allows excessive air infiltration, primarily due to gaps and cracks, resulting in insufficient airtightness. Consequently, this is a threat as this may increase the risk of water ingress and moisture accumulation within these openings, potentially leading to dampness and mold growth, both of which pose health concerns.

To solve this issue, using silicone sealant to seal any visible gaps or cracks in the envelope is a good solution. Also, windows, are a important part of the envelope, contribute to heat loss through air movement. Reducing the overall window area (particularly the openable bit) could help reduce unnecessary air entry and reduce over-ventilation.

Mechanical Ventilation

According to LETI guidelines for small-scale housing, a mechanical ventilation with heat recovery (MVHR) system operating at 90% efficiency is recommended. However, based on previous calculations and Part L benchmarks, the bathroom is under-ventilated while the kitchen is over-ventilated.The bathroom's mechanical fan provides insufficient airflow, leading to air pollutant buildup and condensation, which can cause poor indoor air quality and mold. Increasing the window size could enhance natural ventilation and improve airflow.

In the kitchen, excessive ventilation from the cooker hood results in heat loss and cold draughts. Improving airtightness—by sealing gaps and reducing the openable window area—would help maintain thermal comfort. Despite LETI not setting specific standards for cooker hoods, a recirculating hood with a carbon filter is a good replacement.

To make ventilation through the building even better, using DCV which is demand controlled ventilation would change airflow based on occupancy and humidity, ensuring efficient, good ventilation aligning with LETI standards.

Energy Usage

	Pre-retrofit calculation	LETI document
Gas boiler 72% efficiency	20,870.094	No fossil fuel heating system
Overall energy consumption	372.91 kWh/m^2 PA	35 kWh/m^2 PA

The current gas boiler emits less carbon than electric alternatives (0.184 vs. 0.205 $kgCO_2$/kWh), but its low efficiency (72%) leads to higher energy use, costs, and emissions. A more sustainable option would be replacing it with a heat pump installed beneath the suspended ground floor.

This aligns with LETI's recommendation for fossil fuel-free heating and hot water systems. The home currently uses 372.91 kWh/m²/year to maintain 21°C, far exceeding LETI's target of 35 kWh/m²/year. This highlights inefficiencies in the building's thermal envelope and ventilation.

The aim of this report is to propose retrofit strategies to reduce energy use and meet LETI performance standards.

Task 3: Improve the existing dwelling to meet the client's requirements. – Post-retrofit calculations

Building fabric- The original double-leaf brick wall has a 20mm cavity, 50mm of mineral wool insulation, and plastered internal surfaces. Steel cavity ties connected the brick leaves and secured the insulation. This build up resulted in quite a high U-value of 0.576 W/m²K.

To improve the thermal performance, various material options and insulation types were looked at using manufacturer website data and unbridged U-value calculations. The retrofit design replaces the inner leaf with the following build-up:

1- Airtec AAC block 1

Material	Conductivity (W/mK)	Material thickness - L - (m)	Resistance - R - (m2K/W)
Brick (outer)	0.600	0.102	0.108
Rso sheltered wall			0.110
Air unventilated	0.25	0.025	0.100
AAC (inner)	0.09	0.100	1.111
Plaster	0.3	0.003	1.112
Rsi (wall)	/	/	0.130
TOTAL			2.671
U-Value (W/m2K)			0.374

2- Rockwool Timber frame slab 2

Material	Conductivity (W/mK)	Material thickness - L - (m)	Resistance - R - (m2K/W)
Rso sheltered wall	/	/	0.110
Timber Frame slab inner	0.03	0.140	4.118
Brick (outer)	0.600	0.102	0.108
Air unventilated	0.25	0.025	0.100
Plaster	0.3	0.003	1.112
Rsi Wall	/	/	0.130
TOTAL			5.678
U-Value (W/m2K)			0.176

3- Kingspan Structurally insulated Panel (SIP)3

Material	Conductivity (W/mK)	Material thickness - L - (m)	Resistance - R - (m2K/W)
Brick (outer)	0.600	0.102	0.108
Rso sheltered wall	/	/	0.110
Air unventilated	0.25	0.025	0.100
OSB/3 (inner)	0.13	0.015	0.115
OSB/3 (inner)	0.13	0.015	0.115
SIPs (inner)	0.02	0.142	5.917
Plaster	0.3	0.003	1.112
Rsi (wall)	/	/	0.130
TOTAL			7.707
U-Value (W/m2K)			0.130

4- Kingspan Insulated plasterboard 4

Material	Conductivity (W/mK)	Material thickness - L - (m)	Resistance - R - (m2K/W)
Brick (outer)	0.600	0.102	0.108
Rso sheltered wall	/	/	0.110
Air unventilated	0.25	0.025	0.100
OSB/3 (inner)	0.02	0.050	2.632
Plasterboard	0.22	0.003	0.014
Rsi (wall)	/	/	0.130
TOTAL			3.094
U-Value (W/m2K)			0.323

Blocks | Thomas Armstrong. [online] 2www.rockwool.com. (n.d.). Timber Frame Slab Insulation A1 Non combustible, sound absorbant | Rockwool. [online]

After researching all these materials, the 2 lowest U-Values were achieved when changing to Rockwool Timber frame slab (0.176W/m2K) or when changing to Kingspan structurally insulated panel (0.130W/m2K). This was also reduced with the help of reduction in the air cavity which helps ventilate properly causing less condensation build up.

1- Rockwool Timber Frame Slab

Layers	Material	Proportion of surface area	Material thickness - L - (m)	Thermal Conductivity-	Thermal Resistance -
1	Brick (outer)	100%	0.102	0.600	0.108
2	Rso sheltered wall	100%	/	/	0.110
3	Rsi wall	100%	/	/	0.130
4	Plywood Sheathing	100%	0.012	0.13	0.092
5	Air cavity ventilated	100%	0.050	0.10	0.500
6	Vapor control layer	100%	/		
7	Plasterboard	100%	0.013	0.250	0.052
8	Plaster	100%	0.030	0.20	0.15
9a	Insulated Slab	87.700%	0.120	0.045	4.211
9b	Timber Frame (softwood)	12.300%	0.120	0.15	1.323
TOTAL wall thickness			0.447		

Two distinct thermal pathways are formed where the insulated slab is fitted between sections of the timber frame, corresponding to layers 9a and 9b.

This build-up incorporates Structural Insulated Panels (SIPs) as the inner leaf, complemented by an extra layer of Kingspan Kooltherm K112 insulation board. The board is fitted between sections of softwood timber framing, resulting in two distinct thermal pathways (identified as layers 9a and 9b).

Area = 97.902

Perimeter = 41.986

$B' = \frac{97.902}{0.5(41.986)} = 4.664$

$d_{et} = d_w + \lambda_g (R_{si} + R_f + R_{se})$

$0.447 + 3.06 (0.170 + 0.099)$

$= 0.943\ m$

$U_f = \frac{2 \times 1.5}{\lambda (4.664) + 0.531}$

P:A ratio = 0.429

$U_f = 1.5 / (0.447 (4.664) + 11.000)$

$= 0.100$

$0.26 \frac{2.5 - 4.5}{2.5 - 2} + 0.23 \times \frac{4.5 - 2}{2.5 - 2}$

$U_{value} = 0.100$

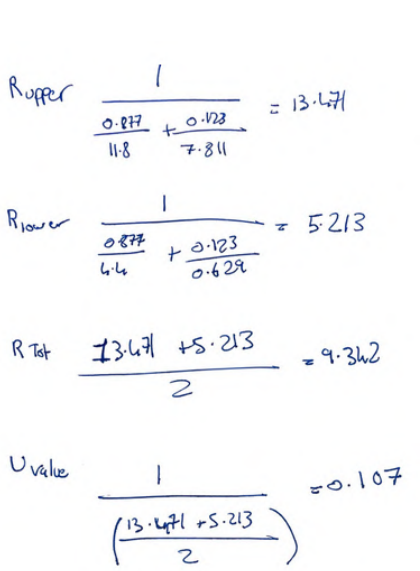
R slab	4.911
R frame	2.023
F slab	0.877
F frame	0.123
R upper	5.660
R lower	4.895
R tot	5.278
U-Value	0.189

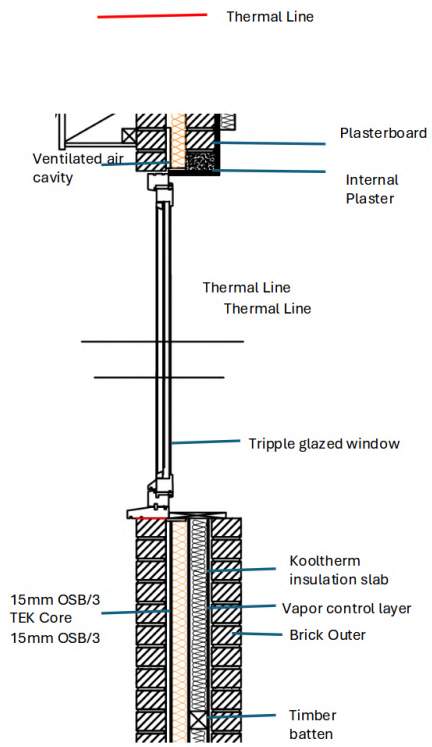
Kingspan SIPs-

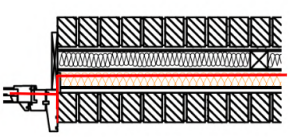
Layers	Material	Proportion of surface area	Material thickness - L - (m)	Thermal Conductivity- (W/mK)	Thermal Resistance - (m2K/W)
1	Brick (outer)	100%	0.102	0.600	0.108
2	Rso sheltered wall	100%	/	/	0.110
3	Rsi wall	100%	/	/	0.130
4	Plywood Sheathing	100%	0.012	0.13	0.092
5	Air cavity ventilated	100%	0.050	0.10	0.500
6	Vapor control layer	100%	/		
7	Plasterboard	100%	0.013	0.250	0.052
8	Plaster	100%	0.030	0.20	0.15
9	OSB/ 3 inner	100%	0.011	0.13	0.085
10	OSB/ 3 inner	100%	0.011	0.13	0.085
11	TEK core (inner)	100%	0.142	0.023	6.174
10a	Kooltherm insulation slab	87.700%	0.088	0.020	4.400
10b	Timber batten	12.300%	0.088	0.14	0.629
TOTAL wall thickness			0.447		

This build-up incorporates Structural Insulated Panels (SIPs) as the inner leaf, complemented by n extra layer of Kingspan Kooltherm K112 insulation board. The board is fitted between sections of softwood timber framing, resulting in two distinct thermal pathways (identified as layers 9a and 9b).

R slab	11.800
R frame	7.811
F slab	0.877
F frame	0.123
R upper	13.471
R lower	5.213
R tot	9.342
U-Value	0.107







Roof

First Pathway

Material	Thickness (m)	Conductivity (W/mK)	Resistance (m2K/W)
Tiled roof with felt			0.200
Rso exposed roof			0.060
Rsi			0.100
Plasterboard	0.012	0.210	0.057
Plaster	0.015	0.3	0.05
Insulation	0.12	0.022	5.456
Insulation between joists	0.074	0.022	3.364
TOTAL	0.221	0.554	9.287

Second pathway-

Material	Thickness (m)	Conductivity (W/mK)	Resistance (m2K/W)
Tiled roof with felt			0.200
Rso exposed roof			0.060
Rsi			0.100
Plasterboard	0.012	0.210	0.057
Plaster	0.015	0.3	0.050
Insulation	0.12	0.022	5.456
Timber Joists	0.070	0.120	0.583
TOTAL	0.217	0.652	6.506
U-value			0.062

Total Area = 97,902

Insulation Area = 400 × 75 = 30,000

Joists Area = 75 × 50 = 3750

Proportion of surface Area:

$\frac{30,000}{97902} = 0.306$

$\frac{3750}{97902} = 0.038$

Upper limit of resistance

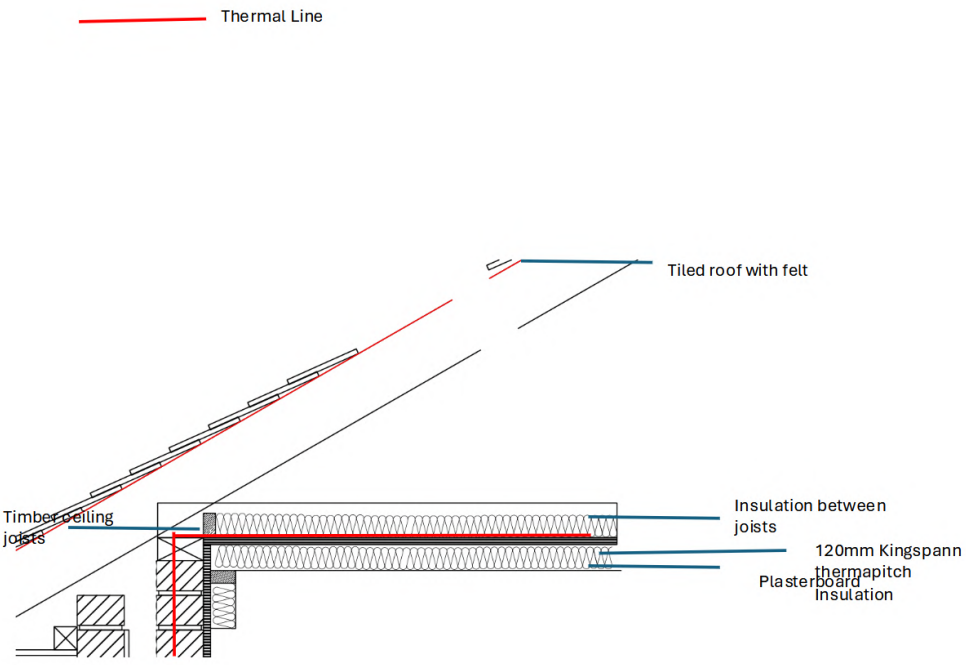
$\frac{1}{\left(\frac{0.306}{9.287}\right) + \left(\frac{0.038}{6.506}\right)} = 25.780$

lower limit of resistance

$\frac{1}{\left(\frac{0.306}{3.364}\right) + \left(\frac{0.038}{0.583}\right)} = 6.404$

R_{tot} $\frac{25.780 + 6.404}{2} = 16.092$

U value $\frac{1}{16.092} = 0.062$



Floor

Material	Thickness (m)	Conductivity (W/mK)	Resistance (m2K/W)
Concrete Cast	0.05	1.52	0.0329
Concrete Cast 2	0.150	1.52	0.0658
Rsi downwards	/	/	0.170
Insulation	0.090	0.02	4.500
TOTAL	0.29	3.06	4.769

To reduce the overall embodied carbon of the building, insulation was installed above the existing oversite concrete rather than opting for a full ground floor replacement. This approach aimed to meet the LETI target floor U-value range of 0.08–0.10 W/m²·K, achieving a value of 0.100 W/m²·K.

An alternative option using the existing 100 mm concrete slab, combined with 90 mm of insulation, would have resulted in a U-value of 0.11 W/m²·K, slightly above the LETI target.

To improve performance, the slab thickness was increased to 150 mm and paired with 90mm of insulation. This revised floor build-up enhances thermal resistance and reduces heat loss. The upgraded floor achieves a U-value of 0.100 W/m²·K, aligning more closely with LETI guidance.

Area = 97.902

Perimeter = 41.986

$B' = \frac{97.902}{0.5(41.986)} = 4.664$

$d_t = d_w + \lambda_g (R_{si} + R_f + R_{se})$

$0.447 + 3.06 (0.170 + 0.099)$

$= 0.943\ m$

$U_f = \frac{2 \times 1.5}{\pi (4.664) + 0.531}$

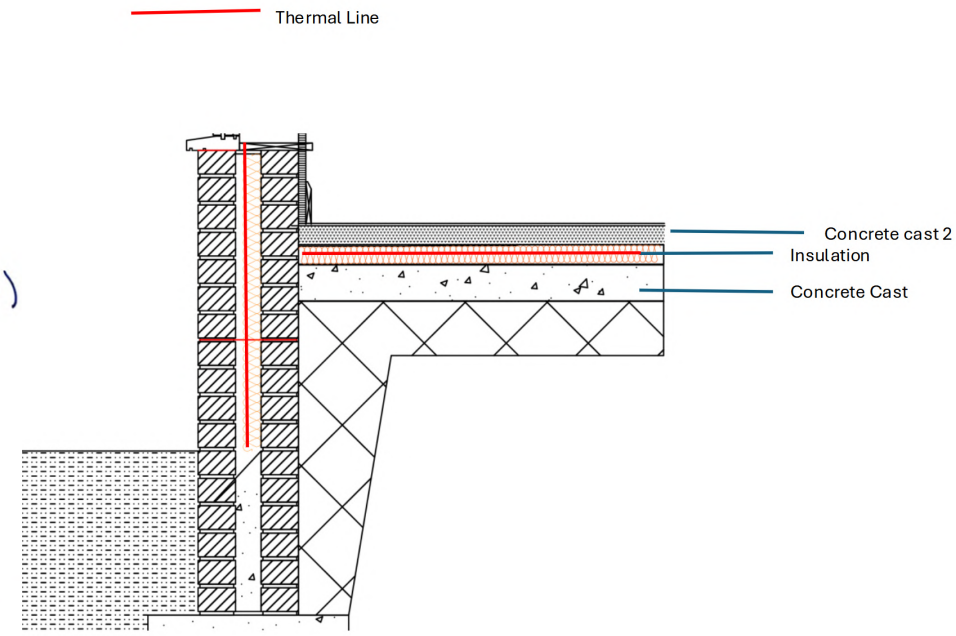
P:A ratio = 0.429

$U_f = 1.5 / (0.447 (4.664) + 11.000)$

$= 0.100$

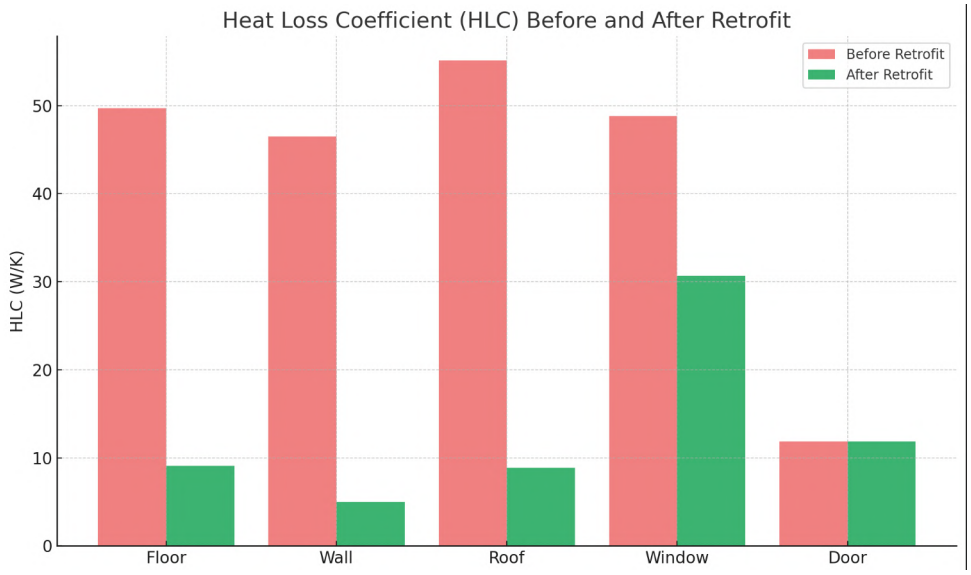
$0.26 \frac{2.5 - 4.5}{2.5 - 2} + 0.23 \times \frac{4.5 - 2}{2.5 - 2}$

$U_{value} = 0.100$



Building Fabric Heat Loss and Annual Energy Demand

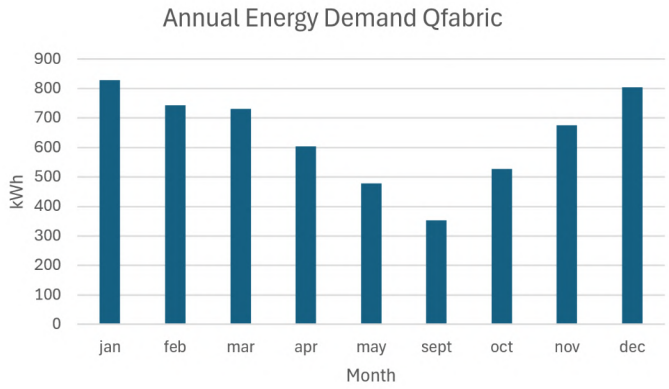
Before retrofit:			
Building element	**Area (m2)**	**U value (m2K/W)**	**HLC (A x U) W/K**
Floor	82.808	0.6	49.685
Wall	80.673	0.576	46.468
Roof	Area of ceiling = 82.808	0.666	55.15
Window	17.443	2.8	48.84
Door	4.75	2.5	11.875
		Total:	**212.018**
After retrofit:			
Building element	**Area (m2)**	**U value (m2K/W)**	**HLC (A x U) W/K**
Floor	82.808	0.11	9.10888
Wall	80.673	0.062	5.001726
Roof	Area of ceiling = 82.808	0.107	8.86
Window	10.959	2.8	30.6852
Door	4.75	2.5	11.875
		Total:	**65.530806**



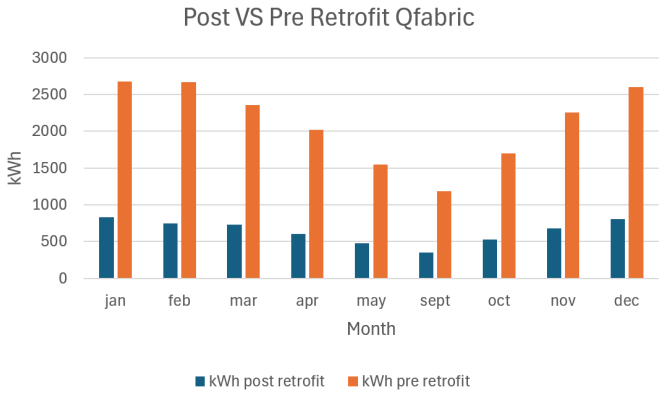
After the retrofit, the total HLC of the building reduced from 212.018 W/K to 65.530806 W/K, showing big improvement in energy efficiency. It had been previously identified that the roof and windows had major room for improvement. The overall improvement in HLC was firstly achieved by reducing the U-values of the floor, walls, and roof quite a lot through better insulation, which lowered heat transfer through these elements by a lot. Also, the total window area was reduced from 17.443 m² to 10.959 m² since windows have a high U-value (2.8), decreasing their size further cut down the overall heat loss. The door's properties remained unchanged. Together, better insulation and a smaller window area led to a much more thermally efficient building.

Annual Energy Demand and CO2 emissions just from fabric heat losses

	HLC	Delta T	HLC x Delta T	Seconds per month	Energy Demand (J)	kWh
jan	65.53	17	1114.01	2678400	2983764384	828.82344
feb	65.53	16.9	1107.457	2419200	2679159974	744.211104
mar	65.53	15	982.95	2678400	2632733280	731.3148
apr	65.53	12.8	838.784	2592000	2174128128	603.92448
may	65.53	9.8	642.194	2678400	1720052410	477.792336
sept	65.53	7.5	491.475	2592000	1273903200	353.862
oct	65.53	10.8	707.724	2678400	1895567962	526.546656
nov	65.53	14.3	937.079	2592000	2428908768	674.69688
dec	65.53	16.5	1081.245	2678400	2896006608	804.44628
				Total:	**20684224714**	**5745.617976**



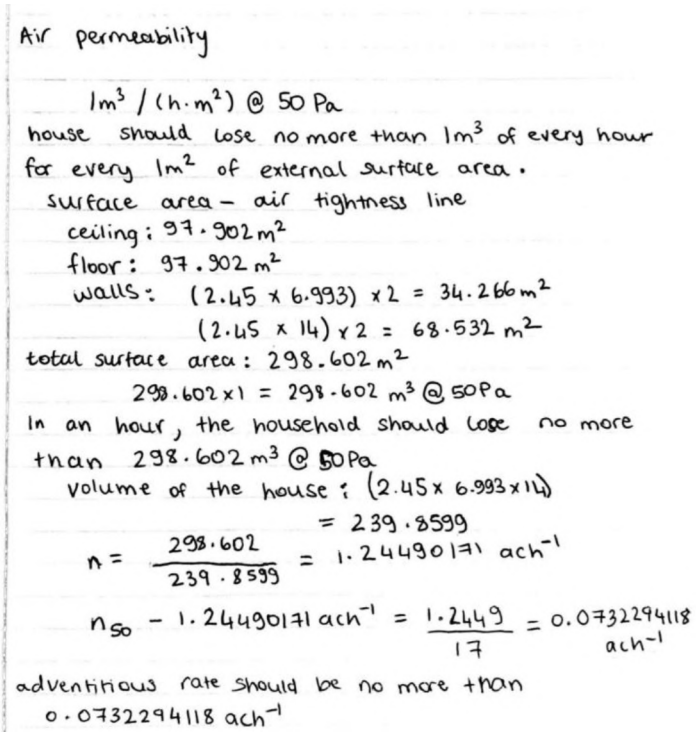
The total energy demand used to be 68,484,997,929.60J which is 19,019.91 kwh pre retrofit. It is now 20,684,224,714J which is 5,745,62kWh. There is a major decrease in the annual energy demand from fabric heat loss due to changes to the construction of the house as explained previously.



This graph compares the building's monthly energy demand (Qfabric) before and after the retrofit. The orange bars are the pre-retrofit energy demand, which is higher throughout the year by a lot, peaking above 2600 kWh in the winter months and staying above 1200 kWh even during the summer. The blue bars are the post-retrofit demand, showing a big reduction across all months, with winter peaks around 800–850 kWh and summer lows around 300–400 kWh. The retrofit measures, such as improved insulation, reduced window areas, and enhanced thermal properties, have drastically lowered the building's energy demand, improving efficiency and reducing heating and cooling loads throughout the year.

Ventilation

Air permeability



By calculating the airflow entering inside the building, the target adventitious infiltration rate is calculated to be $0.07323ach^{-1}$. This rate should be achieved by ensuring minimal gaps or openings during construction, through the usage of sealants. Keeping this level of air tightness will not only reduce heat loss through unintended gaps but also help prevent cold air from entering the building. As a result, less thermal energy will be required from the heating system, thereby lowering overall energy consumption.

Natural Ventilation

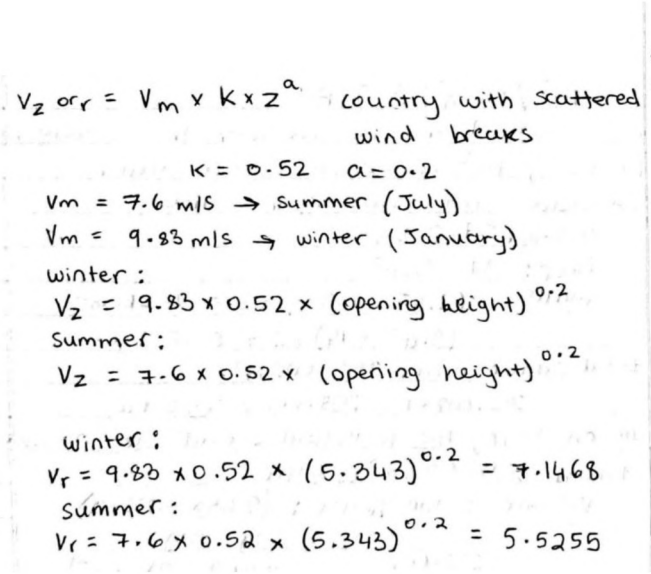
Summer

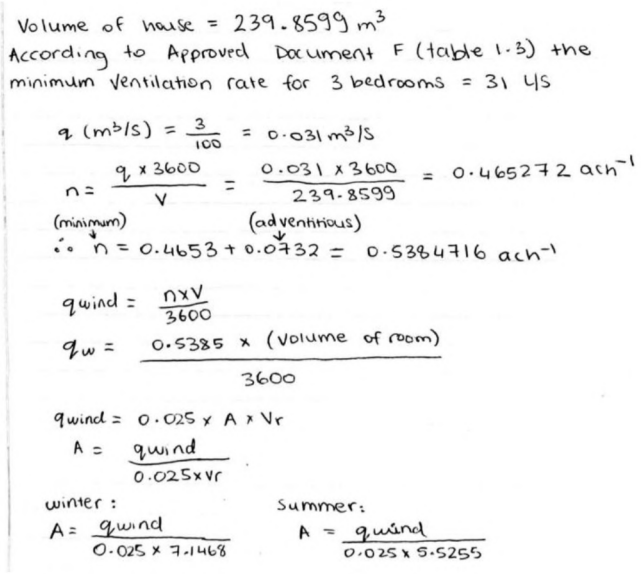
Windows in each room	z – height of opening (m)	Vz	Area of room (m2)	Volume of room (m3)	Qw m3/s	Aw tot	Height of window(m)	Width of openable window (mm)
South – Window 1 Bedroom 1	1.743	4.416	13.494	32.04825	0.00479	0.034676	1.2	17.9
East – Window 2 Bedroom 2	0.825	3.803	9.4746	22.50218	0.00337	0.024396	2.118	12.6
East – Window 3 W.C.	1.825	4.457	1.3687	3.250663	0.00049	0.003547	1.075	15.3
East – Window 4 Living room	0.825	3.803	22.6227	53.72891	0.00804	0.058203	2.118	30.1
West – Window 5 Kitchen	1.825	4.457	12.3913	29.42934	0.0044	0.031852	1.075	16.5
West – Window 6 Bathroom	1.825	4.457	6.0186	14.29418	0.00214	0.015492	1.075	16.6
West – Window 7 Bedroom 3	1.825	4.457	6.3786	15.14918	0.00227	0.016433	1.075	17.6

Winter

Windows in each room	z – height of opening (m)	Vz	Area of room (m2)	Volume of room (m3)	Qw m3/s	Aw tot	Height of window (m)	Width of openable window (mm)
South – Window 1 Bedroom 1	1.743	5.712	13.494	32.04825	0.0048	0.02681	1.2	13.9
East – Window 2 Bedroom 2	0.825	4.919	9.4746	22.50218	0.0034	0.01886	2.118	9.8
East – Window 3 W.C.	1.825	5.765	1.3687	3.250663	0.0005	0.00274	1.075	11.8
East – Window 4 Living room	0.825	4.919	22.623	53.72891	0.0080	0.04499	2.118	23.3
West – Window 5 Kitchen	1.825	5.765	12.391	29.42934	0.0044	0.02463	1.075	12.7
West – Window 6 Bathroom	1.825	5.765	6.0186	14.29418	0.0021	0.01198	1.075	12.9
West – Window 7 Bedroom 3	1.825	5.765	6.3786	15.14918	0.0023	0.01271	1.075	13.6

"Climate & Weather Averages in East Midlands, England, United Kingdom." Www.timeanddate.com, www.timeanddate.com/weather/@11591952/climate.





Single-sided ventilation was applied to these rooms, as each room contains only one window. The calculations identify the required effective opening area (i.e., how wide a window should open) to achieve an air change rate of 0.5385 ach-1, ensuring optimal airflow throughout the house. During the winter, the effective area is reduced to reduce heat loss, because opening the windows too much would allow warm air to leave. In contrast, the effective area is increased in summer to promote greater inflow of cooler air and enhance ventilation.

Mechanical Ventilation

Mechanical Ventilation Heat Recovery (MVHR) System

Lo - carbon MVVR ⇒ 90% efficiency (whole house)
Extract rate : 31 L/S
Duration : 20 hours → 72000s
Total extract = 31 x 7200 = 2232000
In one day, 2232000 / (60x60x24) = 25.83
Doc F → 0.3 L/s TFA for 3 bedroom house
Area of house = 86.658 m^2
∴ 0.3 x 86.658 = 25.997 L/s
Hence when the MVHR system is on for 20 hours
the house is sufficiently ventilated as 25.83 ≈ 25.997 L/S
$n = \frac{q \times 3600}{V}$ $q \rightarrow {}^{1}/_{1000}$ m^3/s

Volume of house = 239.86 m^3
q = 25.83 / 1000 = 0.02583
$n = \frac{0.02583 \times 3600}{239.86} = 0.387676\ ach^{-1}$

HLC = 1/3 x n x V
But the system is 90% efficient
∴ HLC = 1/3 x (1-0.9) x n x V
HLC = 1/3 x (1-0.9) x 0.387676 x 239.86
HLC = 3.099599 W/K



The Vent-Axia13 Lo-Carbon MVHR unit (“MVHR Filters - Mark Heat Recovery Systems”) is used to provide whole-house ventilation and has a flow rate of 31 L/s, complying with ADF Table 1.3. To meet the needed ventilation demand of 25.997 L/s, the system operates for 20 hours per day. Based on this, the extract rate achieved is 25.83 L/s, showing that the property is ventilated enough. As a result, the MVHR system effectively removes indoor air pollutants and prevents condensation, reducing the risk of mold growth, which could otherwise pose health risks to occupants. Also, the unit has 90% heat recovery, meaning it meets LETI guidelines. This means that heat from the extracted air is reused, making the system's energy efficiet. The combined effect of the fan's air change rate and the adventitious air infiltration rate of 0.3877 ACH results in a Heat Loss Coefficient (HLC) of 3.0996 W/K. This HLC value highlights the system’s ability to minimize energy consumption, supporting reduced carbon emissions and improved overall sustainability.

“MVHR Filters - Mark Heat Recovery Systems.” Mark Heat Recovery Systems, 13 Mar. 2024, www.markheatrecoverysystems.co.uk/products/mvhr-filters/. Accessed 24 Apr. 2025.

Kitchen Cooker Hood

Kitchen cooker hood
CDA EVQ71BL 70cm circular Bevelled Glass Hood
Extract rate of up to 579 m^3/h → 161 L/s
Energy rating : A++
Speed setting → "Speed 2" → 378 m^3/h → 105 L/s
Duration = 50 min → 50 x 60 = 3000 s
Total extraction 105 x 3000 = 315,000 L
In one day, 315 000 / (60x60x24) = 3.645833 L/s
$n = \frac{q \times 3600}{V}$ $q \rightarrow {}^{1}/_{1000}$ m^3/s

Volume of kitchen = 30.359 m^3
$q = \frac{3.8458}{1000} = 0.0038458$ $n = \frac{0.0038458 \times 3600}{30.359}$
∴ n = 0.4560387 ach^{-1}
Doc F → 0.3 L/s TFA for 3 bedroom house
Area of kitchen = 12.3913 m^2
0.3 x 12.3913 = 3.71739 L/s
Since 3.84583 ≈ 3.71739 L/s kitchen is sufficiently ventilated
HLC = 1/3 x n x V
n = 0.4560387 + 0.07322944118
n = 0.529 ach^{-1}
HLC = 1/3 x 0.529 x 30.359 = 5.3533 W/K



The CDA EVQ71BL 70cm Circular Bevelled Glass cooker hood (“Cda EVQ71BL 70cm Circular Bevelled Glass Hood 579m3hr”) is a recirculating system with a maximum flow rate of 161 L/s, exceeding the ADF minimum requirement of 60 L/s. When operated for 50 minutes daily, it delivers an average extract rate of 3.845 L/s. Equipped with a charcoal filter, it effectively removes air pollutants. Compared to the Document F requirement of 3.717 L/s, the kitchen is deemed sufficiently ventilated, indicating no excessive draughts and maintaining occupant comfort. However, the hood contributes to a relatively high Heat Loss Coefficient (HLC) of 5.3533 W/K, significantly above that of the bathroom. This cannot be mitigated by heat recovery, as cooker hoods must expel air containing grease and odors and therefore cannot recirculate it to the same extent as MVHR systems.

“Cda EVQ71BL 70cm Circular Bevelled Glass Hood 579m3hr.” Appliances UK, 2025, www.appliancesuk.co.uk/Cda-EVQ7-70cm-Circular-Bevelled-Glass-Hood-760m3hr. Accessed 24 Apr. 2025.

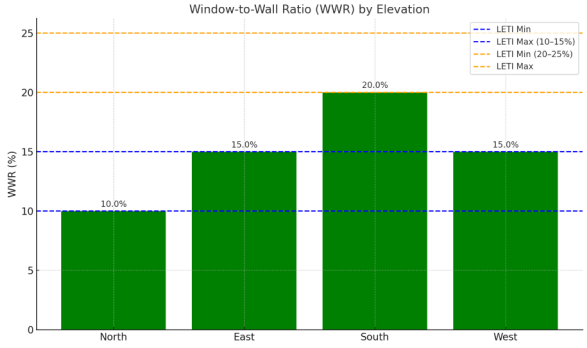
New window over wall area and dimensions of new windows

Elevation	Wall Area (m^2)	Window	Size (W x H in m)	Window Area (m^2)	Total Window Area per Elevation (m^2)	WWR (%)	LETI Target	Complies?
North	17.133	Window 1	1.6 × 1.07	1.713	1.713	10	10-15%	YES
East	22.067	Window 2	2.118 × 0.395	0.836	2.21	10	10–15%	YES
		Window 3	1.075 x 0.5	0.538				
		Window 4	2.118 × 0.395	0.836				
South	14.943	Window 5	1.547 × 1.932	2.989	2.989	20	20-25%	YES
West	26.98	Window 6	1.075 × 2.2	2.365	4.047	15	10-15%	YES
		Window 7	1.075 × 0.782	0.841				
		Window 8	1.075 × 0.782	0.841				

In **blue** is the changed windows to fit the LETI criteria of each elevation. After careful consideration and re calculating new areas, all elevations fall within the LETI target. The North elevation used to have no windows, hence no natural daylight coming from the facade, the total window area for the North elevation used to be 0m2 and it is now 1.713m2, with a WWR % (window over wall ratio) of 10%, complying with the 10-15% suggested by LETI. The East elevation was way over the LETI suggestion of 10-15% WWR as it was 44.673%, causing excessive heat conduction through the window and excessive ventilation. The new windows ensure that the total window area is only 2.21m2, 10% WWR. This is a significant improvement despite being a big change to the design. The South elevation was just under the LETI suggestion as it previously had a WWR of 18.216%, it now complies with LETI and has a WWR of 20% after slightly increasing each window size. Increasing the size of every single window on the South elevation rather than significantly increasing only one window allows for equal distribution of sunlight and prevents one window from causing more overheating than the others. For the West elevation, each window has decreased slightly in size to fall under the 10-15% WWR target by LETI. Going from 18.328% WWR to only 15%.

New window solution by: https://www.saint-gobain-glass.co.uk/product/planitherm-total/

Triple glazed (low-E, soft coat) glazing solution



This graph shows the Window-to-Wall Ratio (WWR) for each elevation of the house compared to LETI guidelines. The WWR values for the North, East, and West elevations fall within the LETI recommended range of 10–15%, ensuring limited heat loss and avoiding overheating. The South elevation, at 20%, aligns with LETI's minimum recommendation for this orientation, which benefits from passive solar gain. The dashed lines indicate LETI’s minimum and maximum targets for different facades, and the graph overall shows that the window design is well-balanced, achieving adequate daylighting while maintaining energy efficiency across all elevations.

New HLC Ventilation and Qventilation

Natural ventilation whole house

HLC = 1/3 x n x V n = 0.5385 ach^{-1}

= 1/3 x 0.5385 x 239.86

= 43.05487

Qv :

Jan	= 43.05487 (21 - 4)	=	731.93279
Feb	= 43.05487 (21 - 4.1)	=	727.627303
Mar	= 43.05487 (21 - 6)	=	645.82305
Apr	= 43.05487 (21 - 8.2)	=	551.102336
May	= 43.05487 (21 - 11.2)	=	421.937726
Sept	= 43.05487 (21 - 13.5)	=	322.911525
Oct	= 43.05487 (21 - 10.2)	=	464.992596
Nov	= 43.05487 (21 - 6.7)	=	615.684641
Dec	= 43.05487 (21 - 4.5)	=	710.405855
	Total Qv	=	5192.417322

MVHR whole house

HLC = 1/3 x n x V x 0.1 (due to 90% efficiency) n = 0.3877 ach^{-1}

HLC = 1/3 x 0.3877 x 239.86 x 0.1

= 3.0996

Qv

Jan	= 3.0996 (21-4)	=	52.6932
Feb	= 3.0996 (21-4.1)	=	52.38324
Mar	= 3.0996 (21-6)	=	46.494
Apr	= 3.0996 (21-8.2)	=	39.67488
May	= 3.0996 (21-11.2)	=	30.37608
Sept	= 3.0996 (21-13.5)	=	23.247
Oct	= 3.0996 (21-10.2)	=	33.47568
Nov	= 3.0996 (21-6.7)	=	44.32428
Dec	= 3.0996 (21-4.5)	=	51.1434
	Total Qv	=	373.81176

Cooker hood ventilation

HLC = 1/3 x n x V n = 0.529 ach^{-1} (total)

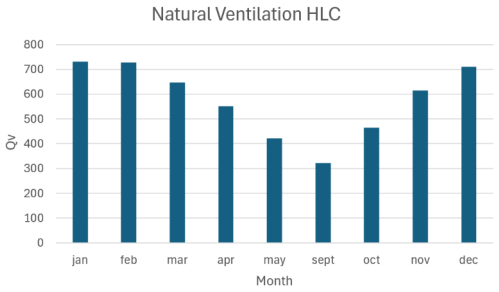
= 1/3 x 0.529 x 30.36

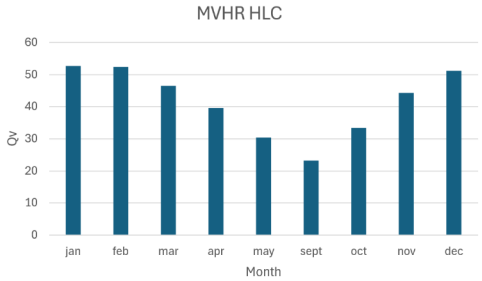
= 5.3533

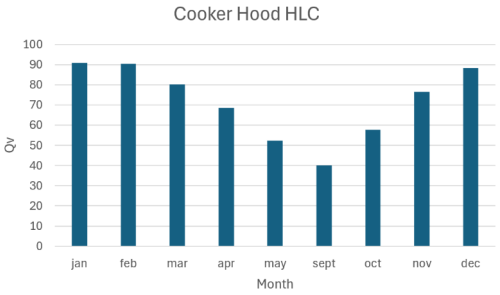
Qv

Jan	= 5.3533 (21-4)	=	91.0061
Feb	= 5.3533 (21-4.1)	=	90.47077
Mar	= 5.3533 (21-6)	=	80.2995
Apr	= 5.3533 (21-8.2)	=	68.52224
May	= 5.3533 (21-11.2)	=	52.46234
Sept	= 5.3533 (21-13.5)	=	40.14975
Oct	= 5.3533 (21-10.2)	=	57.81564
Nov	= 5.3533 (21-6.7)	=	76.55219
Dec	= 5.3533 (21-4.5)	=	88.32945
	Total Qv	=	645.60798

Total ventilation HLC is 51.50777 W/K







Thermal energy loss from natural and mechanical ventilation is calculated based on changes in outdoor temperatures, as these variations impact the temperature difference, leading to monthly fluctuations in heat loss.

Natural Ventilation Losses: These refer to the heat lost through gaps and openings in the building's structure.

Mechanical Ventilation Losses: These capture the heat lost through the operation of systems like the bathroom extractor fan and cooker hood, which remove warm indoor air.

Previously, the HLC was 110.33487 W/K, but it has now reduced to 51.5077 W/K which is a significant improvement resulting from enhanced ventilation systems and the installation of a MVHR system. This indicates that the house is now more energy efficient, with lower heating costs and improved indoor comfort.

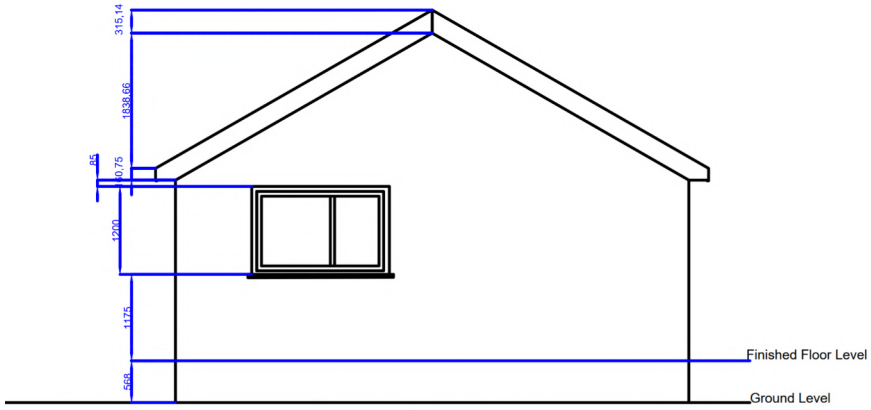
Energy per month and Annual energy demand per month due to ventilation losses

	Natural ventilation Qv	MVHR Qv	Cooker Hood Qv	SUM	Seconds per month	Energy Demand (J)	kWh
jan	731.93279	52.6932	91.0061	875.63209	2678400	2345292990	651.470275
feb	727.627303	52.38324	90.47077	870.481313	2419200	2105868392	584.9634423
mar	645.82305	46.494	80.2995	772.61655	2678400	2069376168	574.8267132
apr	551.102336	39.67488	68.52224	659.299456	2592000	1708904190	474.6956083
may	421.937726	30.37608	52.46234	504.776146	2678400	1351992429	375.5534526
sept	322.911525	23.247	40.14975	386.308275	2592000	1001311049	278.141958
oct	464.992596	33.47568	57.81564	556.283916	2678400	1489950841	413.8752335
nov	615.684641	44.32428	76.55219	736.561111	2592000	1909166400	530.3239999
dec	710.405355	51.1434	88.32945	849.878205	2678400	2276313784	632.3093845
					Annual Energy Demand (kWh)		**4516.160067**

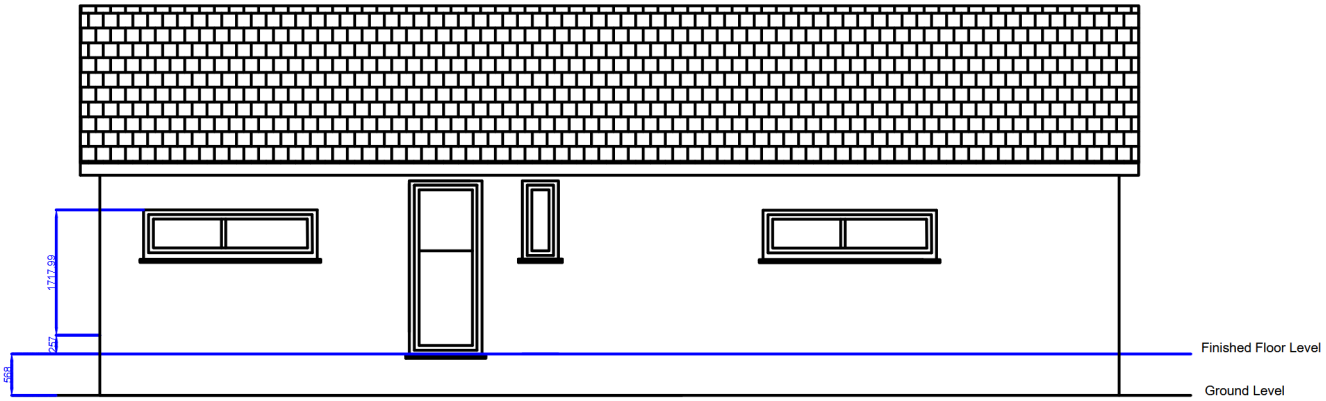
Monthly energy loss is calculated based on Qnatural ventilation and Qmechanical ventilation, excluding the summer months (June, July, and August) when heating is not required and energy losses do not need to be replaced. This calculation reflects the energy lost through ventilation every second over the course of each month, which must be replenished to maintain a comfortable indoor temperature of 21°C.

The annual energy demand was originally 9,674.25 kWh and has now reduced to 4,516.16 kWh, marking a significant decrease. This indicates that the building is now far more energy efficient, resulting in lower heating costs, reduced environmental impact, and improved indoor comfort.

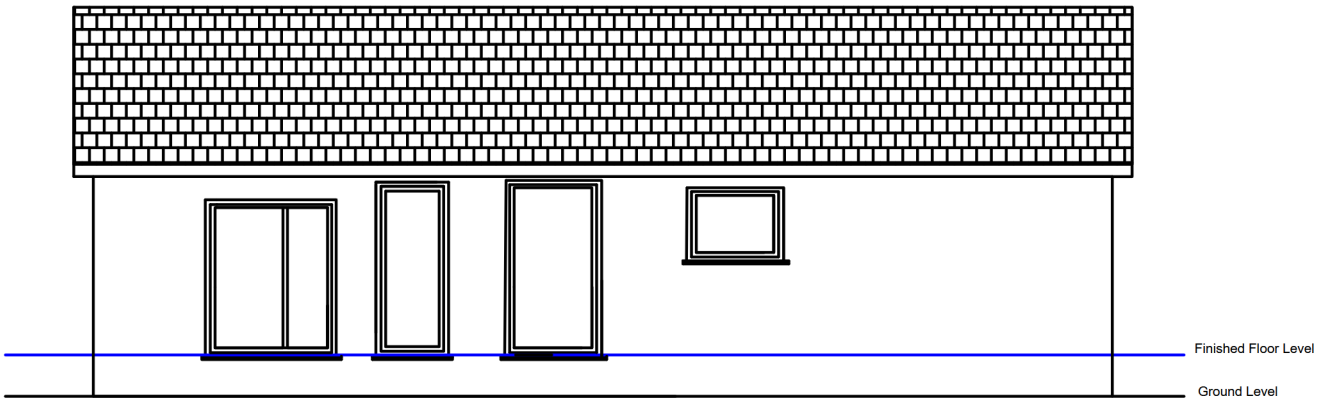
New elevations



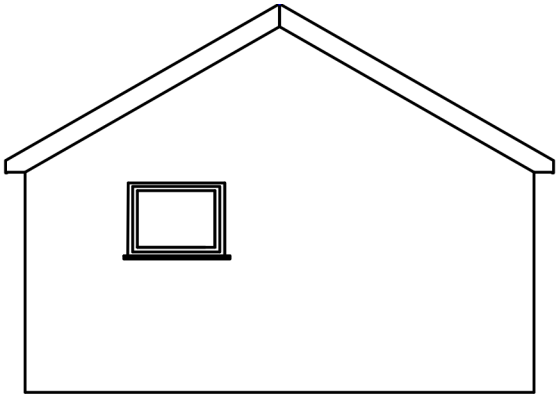
South Elevation



East Elevation



West Elevation



North Elevation

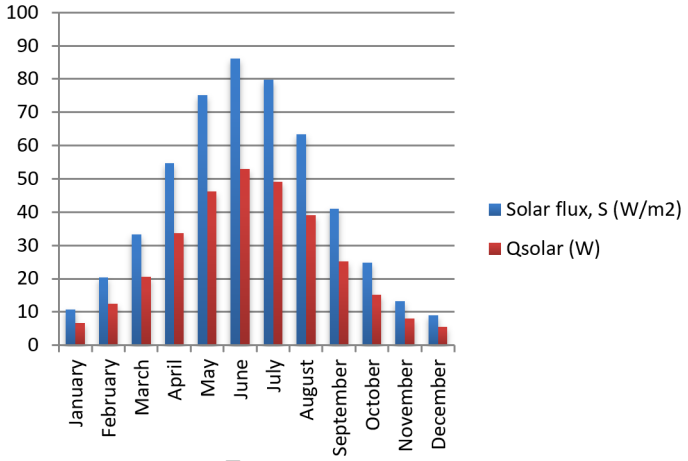
Gains

Solar Gains

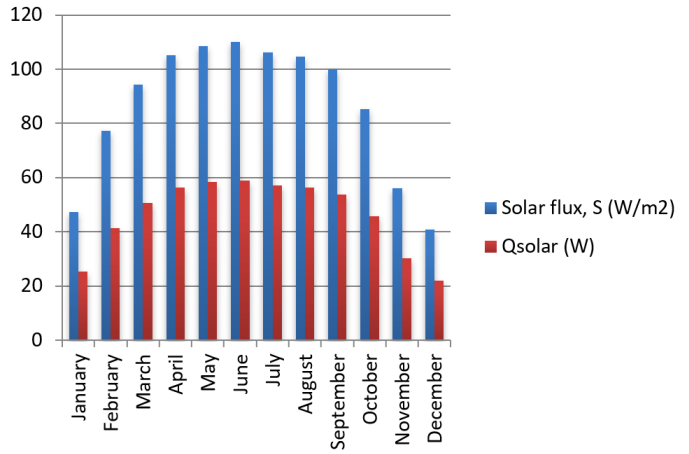
	North, 0		East, 90		South,180		West, 270	
Area of openings (m2)	1.713		2.21		2.989		4.047	
	Solar Flux	Q	Solar Flux	Q	Solar Flux	Q	Solar Flux	Q
jan	10.72641674	6.598229759	19.87256466	15.77108591	47.3233189	25.3972398	19.87256466	28.88035506
feb	20.35879704	12.5234758	38.51867736	30.56884606	77.1832044	41.42229236	38.51867736	55.97833484
mar	33.30872704	20.48947373	61.56524736	48.85885752	94.2460544	50.57949653	61.56524736	89.47140108
apr	54.639576	33.61089589	91.409784	72.54381009	105.11436	56.4122439	91.409784	132.8438006
may	75.2159916	46.26823721	111.2196844	88.26516496	108.549926	58.25602611	111.2196844	161.6330871
june	86.18278656	53.01433281	117.562295	93.29873053	109.9458816	59.00520051	117.562295	170.8506617
july	79.8504485	49.11906914	112.3991865	89.2012308	106.2967225	57.04678823	112.3991865	163.3472312
aug	63.471912	39.04400405	100.108008	79.44681634	104.69172	56.1854236	100.108008	145.4847356
sept	41.085141	25.27304379	73.603769	58.41276072	99.990985	53.66265688	73.603769	106.9667161
oct	24.81434852	15.26425616	46.90850868	37.22710848	85.2918322	45.77398978	46.90850868	68.1710896
nov	13.21801668	8.13090831	24.70675812	19.60755502	56.0693298	30.09100477	24.70675812	35.90578062
dec	8.94448464	5.502095076	16.39290576	13.00959033	40.8904104	21.94485896	16.39290576	23.82344438

The total solar gains used to be 2,405.15W pre retrofit. It is now 703.67W due to the switch from single glazing to triple glazing and the reduction of window areas. These totals exclude the summer months June, July and August as heating isn't required during those months. There is a significant improvement.

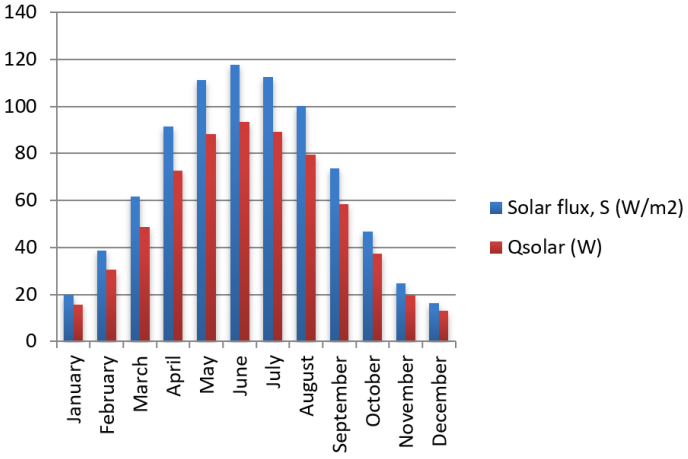
North



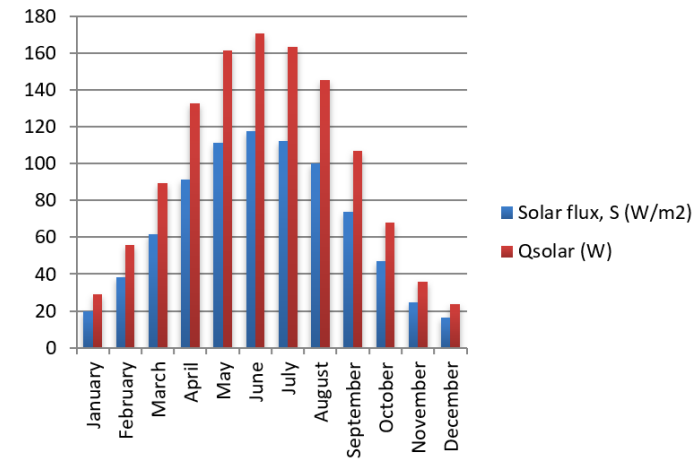
South



East



West



A partial obstruction of the south facade reduces the amount of sunlight reaching south-facing windows, diminishing their role in providing natural heating and consistent lighting, especially during winter. As a result, east-facing windows become more important for delivering bright morning light and passive heating early in the day, while west-facing windows contribute strong afternoon sunlight, which may cause overheating in summer if not properly shaded. North-facing windows continue to receive mostly indirect light, keeping rooms cooler and dimmer year-round. Overall, the obstruction on the south side shifts greater reliance onto the east and west windows, potentially leading to more uneven temperature and lighting conditions throughout the house.

Occupancy schedule

Activity	Room where it takes place	Parent 1	Time spent in each room (min)	Parent 2	Time spent in each room (min)
Waking up	Bedroom	06:30	515 (8.5h)	06:30	515 (8.5h)
Shower	Bathroom	6:30-6:45	15	6:30-6:45	15
Breakfast	Kitchen	6:45-7:00	15	6:45-7:00	15
Leave for work/school	/	7:00-17:00	/	7:00-17:00	/
Do HW with kids	Living room	17:00-18:00	120	17:00-18:00	120
Chores	Kitchen	18:00-19:00	60	18:00-19:00	60
Dinner	Kitchen	19:00 - 20:00	60	19:00 - 20:00	60
Reads/Watches TV	Living Room	20:00-22:00	60	20:00-22:00	60
Sleep	Bedroom	22:00-6:30	/	22:00-6:30	/
Activity	**Room where it takes place**	**Child 1**	**Time spent in each room (min)**	**Child 2**	**Time spent in each room (min)**
Waking up	Bedroom	6:55-7:00	5	6:55-7:00	5
Shower	Bathroom	7:00-7:15	15	7:00-7:15	15
Breakfast	Kitchen	7:15-7:45	30	7:15-7:45	30
Leave for school	/	7:45-16:00	/	7:45-16:00	/
Do HW	Living room	16:00-18:00	120	16:00-18:00	120
Free Time	Bedroom	18:00-19:00	60	18:00-19:00	60
Dinner	Kitchen	19:00 - 20:00	60	19:00 - 20:00	60
Sleep	Bedroom	20:00	/	20:00	/

We have retained the original occupancy schedule.

Lighting Gains

Room	No of fixings	Type	Rating (W)	Time in use (s) per day	Energy (J)	kWh (Heating season)	1 second snapshot (W)	Time in use (s) per day	Energy (J)	kWh (Cooling season)	1 second snapshot (W)
Living room	5	8W LED (E27)	40	14,400	576,000	0.16	40	7,200	288,000	0.080	20
Bedroom 1	5	6W LED (E27)	30	10,800	324,000	0.09	30	7,200	216,000	0.060	20
Bedroom 2	1	6W LED (E27)	6	10,800	64,800	0.02	6	7,200	43,200	0.012	4
Bedroom 3	1	6W LED (E27)	6	10,800	64,800	0.02	6	7,200	43,200	0.012	4
Kitchen	6	5W LED (GU10)	30	14,400	432,000	0.12	30	5,400	162,000	0.045	11.25
Bathroom	4	5W LED (IP44)	20	7,200	144,000	0.04	20	5,400	108,000	0.030	15
W.C.	1	4W LED (IP44)	4	3,600	14,400	0.004	4	1,800	7,200	0.002	2
Hallway	3	4W LED (E14 or GU10)	12	7,200	86,400	0.024	12	3,600	43,200	0.012	6
Total	**26**		**148**	**82,800**	**1,706,400**	**0.47**	**178**	**45,000**	**911,800**	**0.250**	**82.25**

In the new lighting system, all previous incandescent and halogen bulbs are replaced with energy-efficient LEDs, reducing overall power consumption by a lot. In the living room, five 8W E27 LED bulbs have replaced four 100W incandescent bulbs, reducing energy use by over 90% . Bedroom 1 has a reduction from 140W to 30W by switching to five 6W LEDs, while Bedrooms 2 and 3 are replaced a 60W bulb with a 6W LED, achieving a 90% energy cut per room. In the kitchen, six 50W bulbs were swapped for six 5W GU10 LEDs, saving 270W daily. The bathroom's has four 50W bulbs that were replaced with 5W IP44-rated LEDs, dropping from 200W to 20W. Also, the W.C. dropped from a 50W bulb to a 4W LED, and the hallway's three 40W bulbs were replaced with 4W LED candle bulbs, reducing the total to only 12W. Overall, these changes led to an around a decrease of over 72% in daily lighting energy consumption, with added benefits including lower electricity bills, longer bulb lifespan, and improved safety through cooler operation and moisture-resistant fixtures in wet areas.

E27 8W LED Bulb (Philips): www.screwfix.com/p/philips-corepro-ledbulb-b22-806lm-8w-5-pack/2937v

E27 6W LED Bulb (Philips) www.screwfix.com/p/lap-gls-led-light-bulb-es-e27-470lm-6-2w-5-pack/1069r

GU10 5W LED Bulb (Amazon Basics): www.amazon.co.uk/AmazonBasics-LED-Light-Bulb-2-Pack/dp/B07X9ZK5DZ

Bathroom/WC IP65 Downlights: www.toolstation.com/aurora-fire-rated-led-downlight/p45387

E14 4W LED (for Hallway): www.screwfix.com/p/lap-candle-led-light-bulb-ses-e14-470lm-4-2w-5-pack/1713r

Appliance Gains

Room	Type	Rating (W)	Time in use (s)	Energy (Joules)
Whole House	MVHR	2.17	72000	156240
Living room	LCD TV	71	10800	766800
	Games Console	100	3600	360000
	Broadband Router	14	86400	1209600
	Tablet charger	10	3600	36000
	Smartphone charger	5	10800	54000
Kitchen	Oven (electric)	2200	1800	3960000
	Hob (induction)	1500	3600	5400000
	Cooker hood	74	3000	222000
	Fridge freezer	22	86400	1900800
	Dishwasher (eco)	1200	3600	4320000
	Microwave	600	1800	1080000
	Kettle	3050	600	1830000
	Toaster	850	600	510000
	Air Fryer	1800	1800	3240000
Bathroom	Hairdryer	1600	600	960000
Bedroom 1	TV	60	7200	432000
	Electric blanket	75	10800	810000
	Smartphone charger	5	10800	54000
Bedroom 2	TV	40	7200	288000
	Games console	100	7200	720000
	Smartphone charger	5	10800	54000
Bedroom 3	Electric blanket	75	10800	810000
	Smartphone charger	5	10800	54000

Total: 29,227,440 J

Mean heat gain from appliances: QA = 29,227,440 / 24 × 60 × 60 = 338.28 W

To improve energy efficiency of the dwelling, we reviewed the previous list of fixed appliances and looked at the devices with the highest energy consumption. In the new table, many of these appliances were replaced with more energy-efficient alternatives. The changes are marked in purple. For example, the gas hob was replaced with a lower-power induction hob, and the electric oven was replaced with a fan oven that uses less energy. Also, eco-friendlier models were chosen for the dishwasher, air fryer, and electric blankets, while the microwave and fridge freezer were also reduced to units with reduced power ratings. Also, the bathroom extract fan has been replaced by a whole house MVHR system that is 90% efficient, greatly enhancing ventilation performance while minimizing energy loss. As a result of these changes, the total daily energy consumption from appliances dropped significantly, from over 38 million joules to approximately 29 million joules. This led to a noticeable reduction in mean daily heat gains, with the QA value falling from 713.81 W to 338.28 W. These adjustments demonstrate how thoughtful appliance selection can meaningfully reduce internal heat gains and overall energy demand in residential settings.

Microwave (600W): Commercial Chef 600W Countertop Microwave: https://www.amazon.com/Commercial-Chef-0-6-Countertop-Microwave/dp/B01N2HEZJG

Dishwasher (1200W, Eco Model): Farberware Countertop Dishwasher, Energy Efficient: https://www.amazon.com/Farberware-Portable-Countertop-Dishwashers-FDW05ASBWHA/dp/B07VRPB7SN

Induction Hob (1500W): Duxtop 1500W Portable Induction Cooktop: https://www.amazon.com/Duxtop-1800-Watt-Induction-Countertop-9600LS/dp/B083V93HJG

Fan Oven (2200W): Kenwood 2200W Fan Oven Element: https://www.amazon.co.uk/Kenwood-Cooking-Element-2200W-KEN547/dp/B01N7NT7LM

Air Fryer (1800W): Nuwave Brio 8-Quart 1800W Digital Air Fryer: https://www.amazon.com/Nuwave-Brio-8-Quart-Digital-Fryer/dp/B07Q1VM53M

Fridge Freezer (Low Energy – Avg. ~22W): Whirlpool 22 cu. ft. Bottom Freezer Refrigerator: https://www.whirlpool.com/kitchen/refrigeration/refrigerators/bottom-freezer/p.22-cu.-ft.-bottom-freezer-refrigerator-wrb322dmbm.html

Electric Blanket (75W): Heating Pad Electric Blanket 75W: https://www.amazon.com/Sable-Electric-Heating-Auto-Off-Safety/dp/B079DLKF73

Metabolic Gains

Bedroom 1				
Occupant	Activity Level (W)	Time Occupied (h)	Time Occupied (s)	Gains (J)
Parent 1	115	8.5	30600	3519000
Parent 2	97.75	8.5	30600	3519000
			Total (J)	7038000
			Total (W)	81.46

Bedroom 2				
Occupant	Activity Level (W)	Time Occupied (h)	Time Occupied (s)	Gains (J)
Child 1	86.25	11	39600	3415500
			Total (J)	3415500
			Total (W)	39.53

Bedroom 3				
Occupant	Activity Level (W)	Time Occupied (h)	Time Occupied (s)	Gains (J)
Child 2	86.25	11	39600	3415500
			Total (J)	3415500
			Total (W)	39.53

Bathroom				
Occupant	Activity Level (W)	Time Occupied (h)	Time Occupied (s)	Gains (J)
Parent 1	115	0.25	900	103500
Parent 2	97.75	0.25	900	87975
Child 1	86.25	0.25	900	77625
Child 2	86.25	0.25	900	77625
			Total (J)	346725
			Total (W)	4.01

Living Room				
Occupant	Activity Level (W)	Time Occupied (h)	Time Occupied (s)	Gains (J)
Parent 1	115	3	10800	1242000
Parent 2	97.75	3	10800	1055700
Child 1	86.25	2	7200	621000
Child 2	86.25	2	7200	621000
			Total (J)	3539700
			Total (W)	40.97

Kitchen				
Occupant	Activity Level (W)	Time Occupied (h)	Time Occupied (s)	Gains (J)
Parent 1	115	2.25	8100	931500
Parent 2	97.75	2.25	8100	791775
Child 1	86.25	1.5	5400	465750
Child 2	86.25	1.5	5400	465750
			Total (J)	2654775
			Total (W)	30.73

Total Metabolic Gains Average over a one second snap shot of the day (W)
Total : 236.23

Metabolic gains have remained the same where we are assuming that they were sitting at rest the whole time, we assumed their activity level as 115W for a male and applied the correction factor of 0.85 for the female (making 97.75W) and 0.75 for the children (making 86.25).

Heat gains (unregulated)

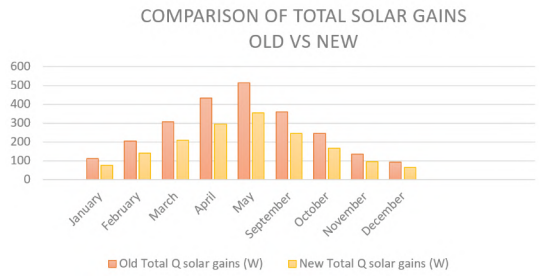
Month	Solar (W)	Lighting (W)	Appliance (W)	Metabolic (W)	Fabric Loss (W)	Ventilation Loss (W)	Total Gains (W)	Total Losses (W)	Losses - Gains (W
Jan	76.64691054	130	338.28	236.23	828.82344	651.470275	781.1569105	1480.293715	699.1368044
Feb	140.4929491	130	338.28	236.23	744.211104	584.9634423	845.0029491	1329.174546	484.1715973
Mar	209.3992289	130	338.28	236.23	731.3148	574.8267132	913.9092289	1306.141513	392.2322843
Apr	295.4107505	130	338.28	236.23	603.92448	474.6956083	999.9207505	1078.620088	78.69933779
May	354.4225154	130	338.28	236.23	477.792336	375.5534526	1058.932515	853.3457886	-205.5867268
Sept	244.3151775	130	338.28	236.23	353.862	278.141958	948.8251775	632.003958	-316.8212195
Oct	166.436444	130	338.28	236.23	526.546656	413.8752335	870.946444	940.4218895	69.47544547
Nov	93.73524871	130	338.28	236.23	674.69688	530.3239999	798.2452487	1205.02088	406.7756312
Dec	64.27998875	130	338.28	236.23	804.44628	632.3093845	768.7899887	1436.755665	667.9656758
TOTAL	**1645.139213**	**1170**	**3044.52**	**2126.07**	**5745.617976**	**4516.160067**	**7985.729213**	**10261.77804**	**2276.04883**

The losses have surpassed gains for every month except May and September, which are closer to summer when higher solar gains and warmer outdoor temperatures reduce heating needs. Losses exceeding gains overall indicates improvements like better ventilation, helping maintain indoor air quality, and controlled heat loss through the building fabric. It also shows that while passive solar gains contribute significantly, additional heating is still needed in colder months. The implementation of a MVHR system for the whole house, replacement of the cooker hood for a better alternative in the kitchen reduce excessive heat gains from ventilation. The re-thinking of building materials for the build up of the roof, floor and walls and addition of insulation lead to a reduction in the u-values and therefore a lower HLC. Replacing light bulbs with lower wattage LEDs allow for excessive heat gains to the house from the bulbs to be diminished. Reducing areas of windows and switching from single glazed to triple glazed minimise heat gain and loss depending on the season. Smarter appliance choices decrease the heat gains. Overall, the building balances comfort, ventilation, and energy efficiency effectively across the seasons.

Comparison Graphs OLD VS NEW - GAINS/LOSSES

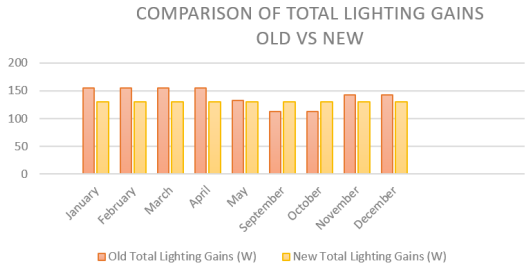
Solar Gains

COMPARISON	Old vs New	
Month	**Old Total Q solar gains (W)**	**New Total Q solar gains (W)**
January	111.1530105	76.64691054
February	205.3973899	140.4929491
March	308.1593809	209.3992289
April	433.4413556	295.4107505
May	513.9972881	354.4225154
September	359.62578	244.3151775
October	244.1165989	166.436444
November	136.2905269	93.73524871
December	92.97279253	64.27998875



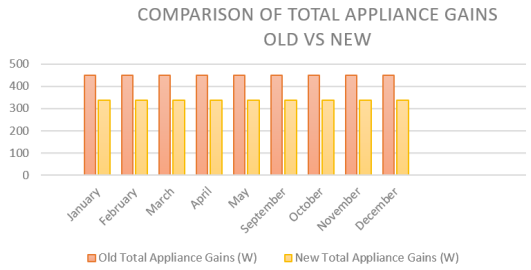
Lighting Gains

COMPARISON	Old vs New	
Month	**Old Total Lighting Gains (W)**	**New Total Lighting Gains (W)**
January	154.58	130
February	154.58	130
March	154.58	130
April	154.58	130
May	132.92	130
September	112.08	130
October	112.08	130
November	142.08	130
December	142.08	130



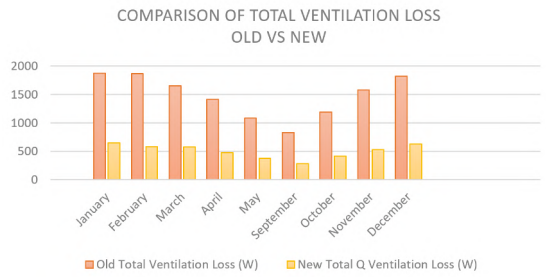
Appliance Gains

COMPARISON	Old vs New	
Month	**Old Total Appliance Gains (W)**	**New Total Appliance Gains (W)**
January	449.23	338.28
February	449.23	338.28
March	449.23	338.28
April	449.23	338.28
May	449.23	338.28
September	449.23	338.28
October	449.23	338.28
November	449.23	338.28
December	449.23	338.28



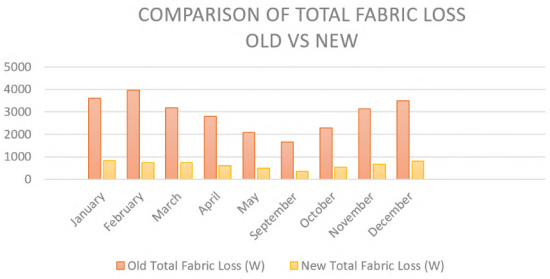
Ventilation loss

COMPARISON	Old vs New	
Month	**Old Total Ventilation Loss (W)**	**New Total Q Ventilation Loss (W)**
January	1875.725806	651.470275
February	1864.702381	584.9634423
March	1655.053763	574.8267132
April	1412.319444	474.6956083
May	1081.303763	375.5534526
September	826.8333333	278.141958
October	1191.639785	413.8752335
November	1577.819444	530.3239999
December	1820.560484	632.3093845



Fabric loss

COMPARISON	Old vs New	
Month	**Old Total Fabric Loss (W)**	**New Total Fabric Loss (W)**
January	3604.301075	828.82344
February	3967.008929	744.211104
March	3172.80914	731.3148
April	2802.819444	603.92448
May	2077.782258	477.792336
September	1643.902778	353.862
October	2289.045699	526.546656
November	3133.069444	674.69688
December	3502.069892	804.44628



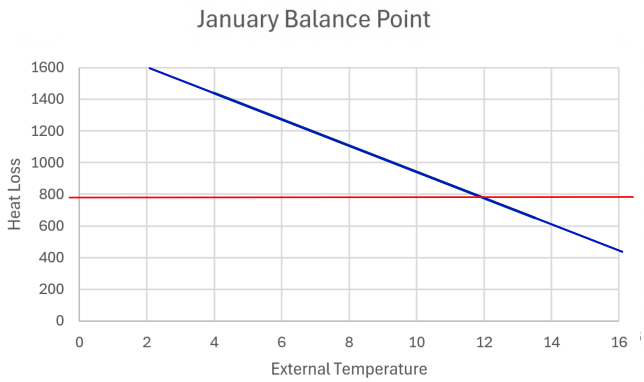
Energy consumption (regulated)

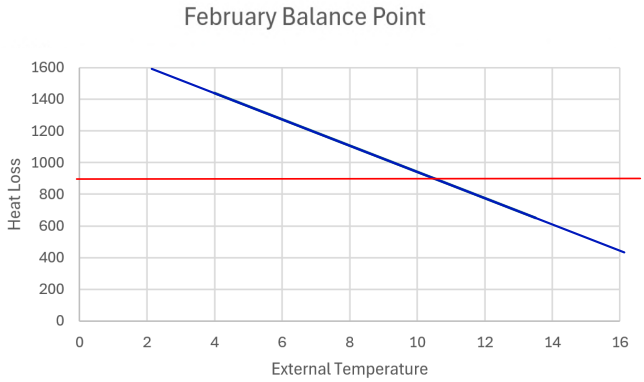
Balance point and degree days:

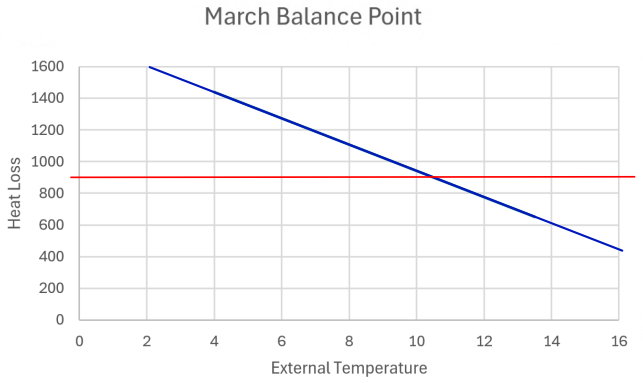
Month	External temperature	Qloss (W)	Qgain (W)	Balance point	Rounded balance point
Jan	4	1480.293715	781.1569105	12.02903205	12
Feb	4.1	1329.174546	845.0029491	10.25607635	10
Mar	6	1306.141513	913.9092289	10.50447689	11
Apr	8.2	1078.620088	999.9207505	9.133926166	9
May	11.2	853.3457886	1058.932515	8.838999396	9
Sept	13.5	632.003958	948.8251775	9.740278061	10
Oct	10.2	940.4218895	870.946444	10.99787042	11
Nov	6.7	1205.02088	798.2452487	11.52721223	12
Dec	4.5	1436.755665	768.7899887	12.17105634	12

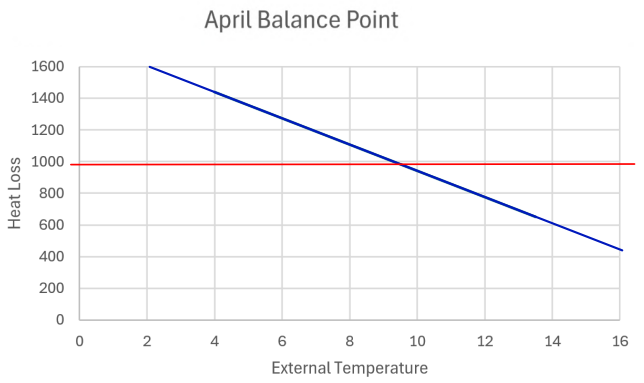
On a graph, a balance point is the point where heat loss intersects with heat gain. This means that there is no excess heat gain or loss and the house is at an internal comfort temperature of 21 C.

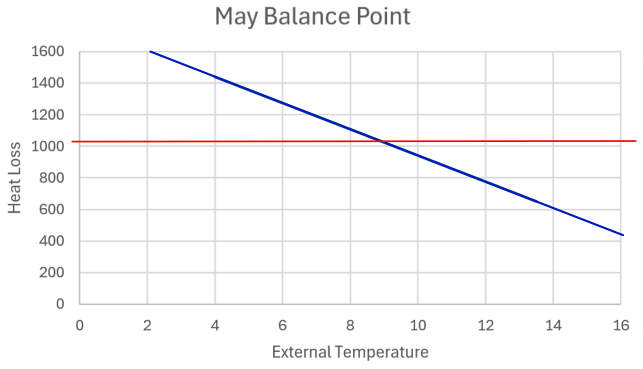
Per month data:

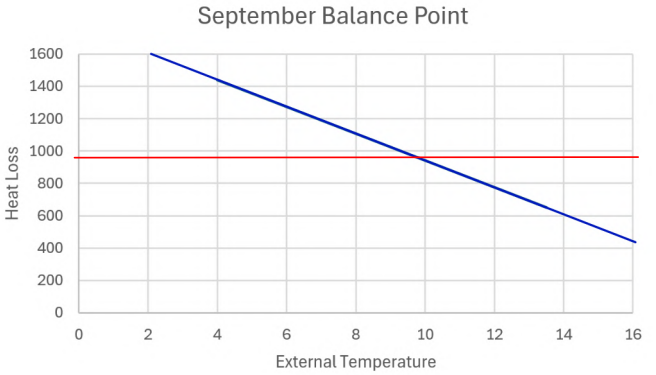


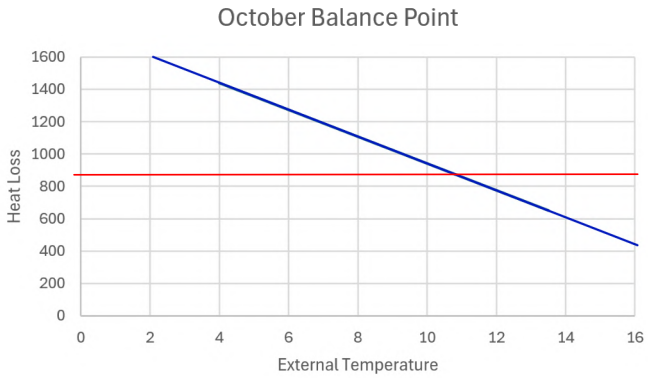


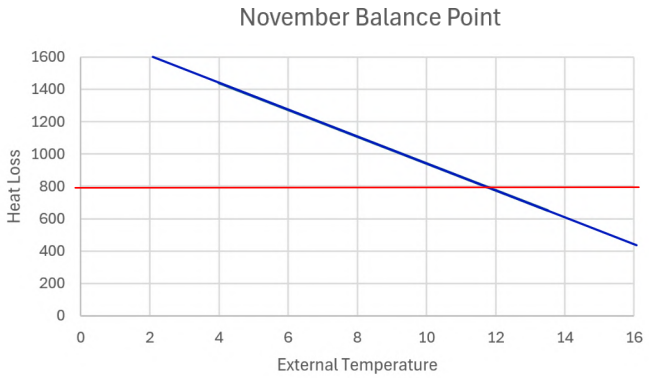


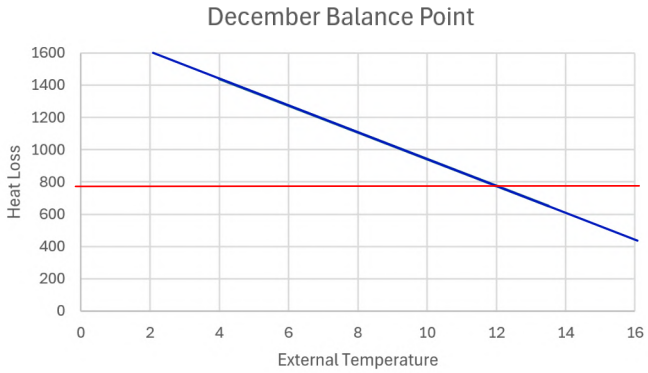












Jan – heat gain = 781.1569 W

formula = $y = -82.9x + 1770.2$

$\therefore 781.1569 = -82.9x + 1770.2$

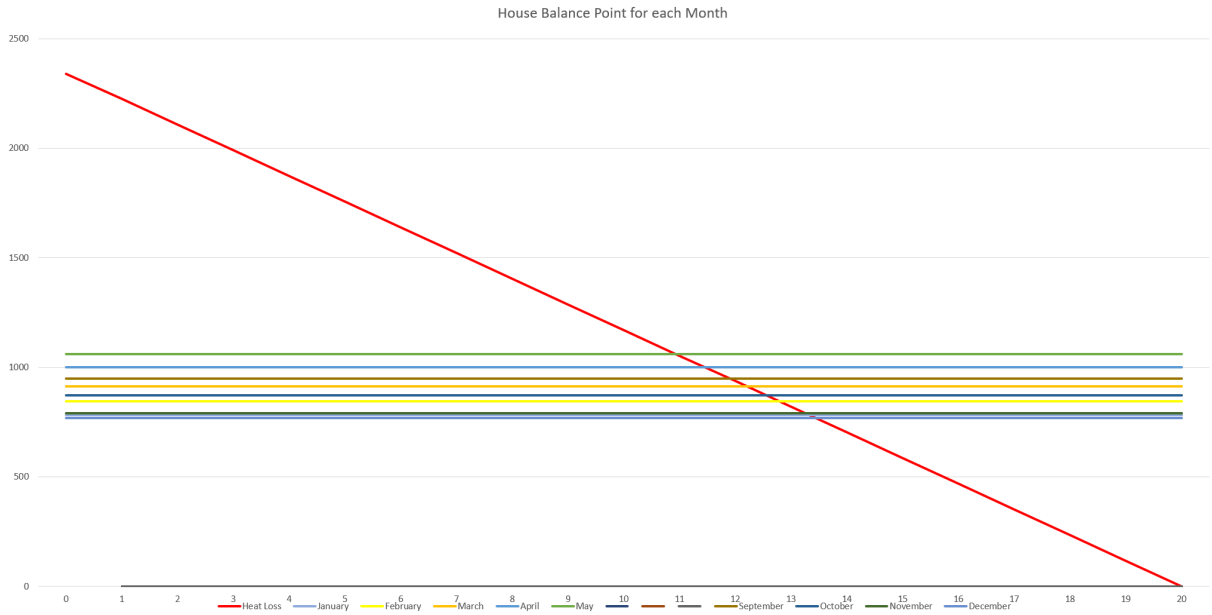
$x = ?$

$x = \frac{781.1569 - 1770.2}{-82.9}$

$x = 11.93 \Rightarrow 12°C$

As illustrated in the following graph, the heat loss trend line is represented by the equation y = -82.9x + 1770.2. The y-values of the horizontal trend lines indicate the monthly heat gains, which can be used to determine the balance points, either by calculation or by simply reading the graph. During the summer months, temperatures are too high for heating to be necessary, so those periods are not included in the calculations.

Balance points throughout the year



The balance point graph shows that colder months, such as January, February, and December, have higher balance points, showing that more heating is required to maintain indoor comfort. In these months, the building loses more heat to the environment and a higher outside temperature is needed before heating is not necessary anymore. In contrast, during hotter months like April, May, and September, the balance points are lower because the natural heat gains from internal sources and solar radiation are enough to maintain comfortable indoor temperature of 21 degrees celcuis with barely to no additional heating. The crossings between the monthly heat gain lines and the heat loss trend line represent the balance points for each month. In general, the graph shows that as outdoor temperatures increase, heat loss decreases, reducing most need for heating, especially during warmer

Degree days

Balance points are used to find degree days, which provide information for the energy needed per month for the building to maintain a comfort temperature of 21℃.This information was calculated on the website “degreedays.net” (DegreeDays) based on balance points and the data from the East midlands’ airport.

Description:	Celsius-based heating degree days with base temperatures at and around 12 C												
Source:	www.degreedays.net												
Accuracy:	No problems detected												
Station:	East Midlands (1.33W,52.83N)												
Station ID:	EGNX												
	(Column titles show the base temperature in Celsius)												
Month starting	HDD 9	HDD 9.5	HDD 10	HDD 10.5	HDD 11	HDD 11.5	HDD 12	HDD 12.5	HDD 13	HDD 13.5	HDD 14	HDD 14.5	HDD 15
01/04/2024	34.6	42.1	50.4	59.5	69	79.4	90.6	102.2	114.3	127	140.2	153.6	167.2
01/05/2024	0.4	1	1.3	3	4.4	7.6	11	15.9	21.1	28.2	35.7	45.5	54.6
01/06/2024	1.5	2.5	3.7	5.7	7.8	10.6	14.2	19	23.5	29.7	36.2	43.8	52.3
01/07/2024	0	0	0	0.3	0.5	1.4	2.3	4.8	6.8	9.7	13.5	17.6	22.7
01/08/2024	0	0.1	0.3	0.8	1.4	2.4	3.2	4.8	6.6	8.9	11.7	16	19.5
01/09/2024	5.3	7.2	9.4	11.9	14.9	18.7	22.9	28.5	34	41.2	48.7	57.6	66.8
01/10/2024	11.7	15	18.4	23.3	28.9	36.4	44.1	54.2	64.4	76.7	89	102.8	116.8
01/11/2024	77.2	87	97	109	120.9	134.2	147.7	161.9	175.9	190.5	205	219.6	234.2
01/12/2024	75.6	87.5	99.9	113.4	126.8	141.1	155.6	170.6	185.9	201.3	216.8	232.3	247.8
01/01/2025	181	196.1	211.4	226.7	242	257.5	273	288.5	304	319.5	335	350.5	366
01/02/2025	120.8	133	145.5	158.1	171.1	184.3	197.6	211.1	224.7	238.4	252.3	266.2	280.2
01/03/2025	75.1	85.3	95.6	107.1	118.2	130.8	143.6	156.8	170.4	184.3	198.5	212.9	227.5

Month	HDD	HLC Loss	Constant	Energy (kWh)
Jan	273	117.038	0.024	766.831469
Feb	145.5	117.038	0.024	408.6958928
Mar	107.1	117.038	0.024	300.833884
Apr	34.6	117.038	0.024	97.18816421
May	0.4	117.038	0.024	1.123562592
Sept	9.4	117.038	0.024	26.40372091
Oct	28.9	117.038	0.024	81.17739727
Nov	134.2	117.038	0.024	376.9552496
Dec	155.6	117.038	0.024	437.0658483
Total				**2496.275189**

energy = power x time

∴ energy (kWh) = $(HLC_F + HLC_V)$ x degree days x 0.024

HLC_F + HLC_V = HLC loss

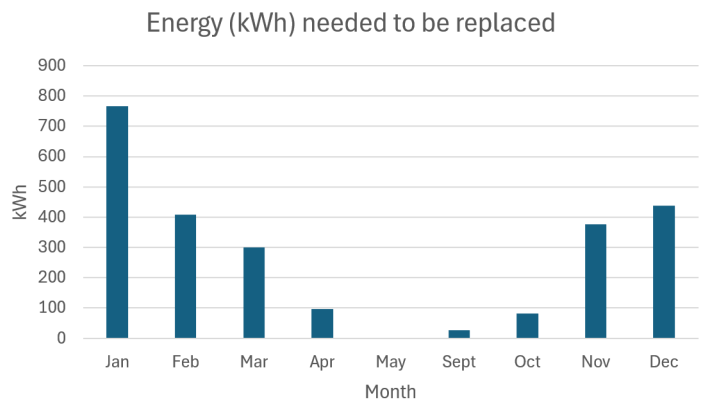
0.024 = constant. $\frac{60 \times 60 \times 24}{360\,000}$

jan → (65.53 + 51.50777) x 273 x 0.024

= 766.831469 kWh

Hence, in January, assuming a balance point of 12℃, 766.831469 kWh of additional energy is needed in order to maintain a comfort temperature in the house, which will be provided by a heating system.

This will allow for the calculation of carbon as a result of this.



Throughout the year, a total of 2,496.275189 kWh will need to be replaced by the heating system. More heating will be needed in colder (winter) months such as January, February, November and December. Conversely, less heating is needed during the transitional (autumn/spring) months of march April may September and October.

DegreeDays. “Heating & Cooling Degree Days - Free Worldwide Data Calculation.” Degreedays.net, 2019, www.degreedays.net/.

Heating system

For the heating system, the boiler is replaced with a wall mounted condensing combination boiler (“EcoTEC plus | Vaillant”) that uses natural gas and has up to 94% ErP. This means the boiler can achieve an efficieny of up to 94% under optimal conditions, does not mean it always operates at 95% efficiency. Therefore for the calculations we can assume it is at 90.1% efficiency since appliance types such as this are usually at 90.1% efficiency.

Month	Energy (kWh)	kWh per m2	Energy consumed by new boiler Kwh	C02
Jan	766.831469	9.260354906	851.0893108	187.2396
Feb	408.6958928	4.935463878	453.6025448	99.79256
Mar	300.833884	3.632908463	333.8888835	73.45555
Apr	97.18816421	1.173656702	107.8669969	23.73074
May	1.123562592	0.013568286	1.247017305	0.274344
Sept	26.40372091	0.318854711	29.30490667	6.447079
Oct	81.17739727	0.980308633	90.0970003	19.82134
Nov	376.9552496	4.552159811	418.3743059	92.04235
Dec	437.0658483	5.278063089	485.0897317	106.7197
Total	**2496.275189**	**30.14533848**	**2770.560698**	**609.5234**

Efficiency	90.10%
Natural Gas Emission	0.22
Floor area m2	82.808

The total CO2 used to be 7848.71 and it is now 608.52, showing significant improvement through the selection of the new boiler.

The total energy required by the house used to be 30,880.17kWh and it is now 2,496.2751kWh. This means 2,496.2751kWh is required to keep the building at 21 degrees celcius. This is a significant improvement coming from new U-values for the construction of the roof, floor and walls, reduction of windows, switching of appliances and lighting and better ventilation systems and cooker hood. All improvements have contributed to this significant decrease.

Cooling:

During summer months, achieving an air change rate of 0.5385 ach^{-1} through natural ventilation could be done by opening the windows wider, and so increasing the effective ventilation area compared to earlier estimates. This means fresh air is allowed to enter the building, helping to remove indoor pollutants and maintain the comfortable indoor temperature of 21 degrees celcius. Given that the UK climate is generally temperate, with relatively mild weather conditions, the addition of air conditioning is quite unnecessary. Also, air conditioning systems contribute to higher energy consumption and electricity demand, meaning there would be an increase in greenhouse gas emissions and a greater overall carbon footprint if there was an air conditioning unit.

“EcoTEC plus | Vaillant.” Vaillant.co.uk, 2024, www.vaillant.co.uk/product-systems/gas-boilers/ecotec-plus/.

Task 5: Summary, benchmark and reflection

Overall, the building's total energy consumption and CO_2 emissions have improved substantially following the retrofit. This improvement is primarily due to reduced heat loss through the building fabric and ventilation systems, achieved by adding insulation to the roof, walls, and floors, and by replacing single-glazed windows with triple glazing throughout. In terms of ventilation, the removal of the old bathroom extractor fan and the installation of a new kitchen cooker hood have further minimized heat loss. The introduction of a Mechanical Ventilation with Heat Recovery (MVHR) system ensures uniform ventilation across the house, preventing over- or under-ventilation in individual rooms. This has significantly enhanced the building's overall energy efficiency, as the energy consumption calculations for the heat pump system account for balance points and degree days. As a result, the building now successfully employs a passive design strategy that aligns with current climate performance standards.

Furthermore, a comparison of the total kWh and carbon emissions per annum between the post-retrofit house and the original structure shows that the building is now over fourteen times more energy efficient. Emissions have been dramatically reduced from approximately 8717.10 $kgCO_2$ per year to 609.52 $kgCO_2$ per year, significantly lessening the home's carbon footprint.According to LETI standards, a home should aim to produce no more than 35 kWh/m^2 per annum; currently, the retrofitted house achieves 30.145 kWh/m^2 per annum. This strong performance is particularly notable given the challenges posed by geometric and thermal bridging within the new building envelope, although all new U-values comply with the recommended LETI thresholds. While window areas were significantly reduced to improve thermal performance, care was taken to preserve key outward views, particularly from important spaces such as the living room, and to maintain high levels of natural daylight, thereby reducing the need for artificial lighting. Additionally, the use of the MVHR system, operating for 20 hours daily, contributed significantly to increased appliance gains. Nevertheless, the house successfully meets the LETI energy consumption target, achieving 30.145 kWh/m^2 per annum (a substantial improvement from the previous figure of 372.91 kWh/m^2 per annum) demonstrating major enhancements in both the building's sustainability and overall energy performance.